Pedro Pathing 3 Tuning Instructions using OnBotJava

Tuning your robot for Pedro Pathing using OnBotJava is very similar to how it is done with Android Studio. It requires loading the Pedro Pathing Quickstart application on the robot instead of the normal FTC application. These instructions will step through how to switch between those applications.

Tuning

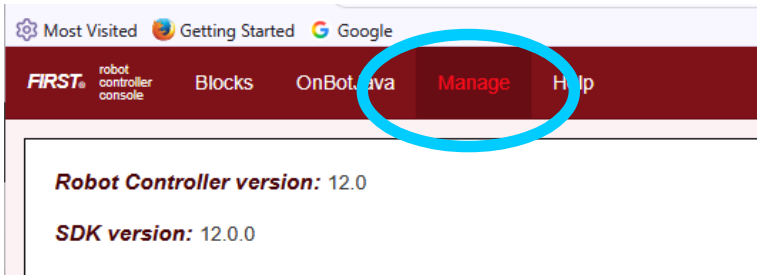
1. Upload the Constants.java file from this folder to OnBotJava
2. Load the Quickstart application on the Control Hub using the instructions below
3. Go to Quickstart Tuning instructions web pages (https://pedropathing.com/docs/pathing/tuning)
4. You do not need to do the installation section since it was done in step 2 above.
5. Skip the "Add the Tuner" sections – they have already been added in the Quickstart app, so there is no Tuning.java file needed in OnBotJava.
6. If the tuner stops with an error (usually null class reference), there is a log file in the same directory as Constants.java called pedroTuningResults.txt. It may list some errors. Also look for number values of -999999.9 – these indicate a parsing error in the Constants.java file.
   a) To update the pedroTuningResults.txt file, you need to reload the file by closing the editor tab for that file and re-opening the file.

After Tuning

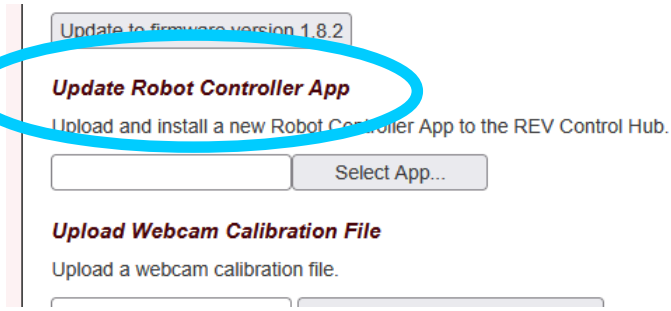
1. Once tuning is done, you need to load the default FTC control hub application. Note that the Constants.java file will not build without errors during the Tuning phase (when Quickstart app is loaded).
2. To load the default FTC control hub application, use either the Rev Hardware Client or download the FTC application from Github and load via the Manage tab. See instructions below.
3. After loading the default application, do the following:
   a) Upload the PedroPathing.jar file
4. You should now be able to build your Constants.java file and any path files you have loaded from the Visualizer.

Loading Quickstart application on the Contol Hub

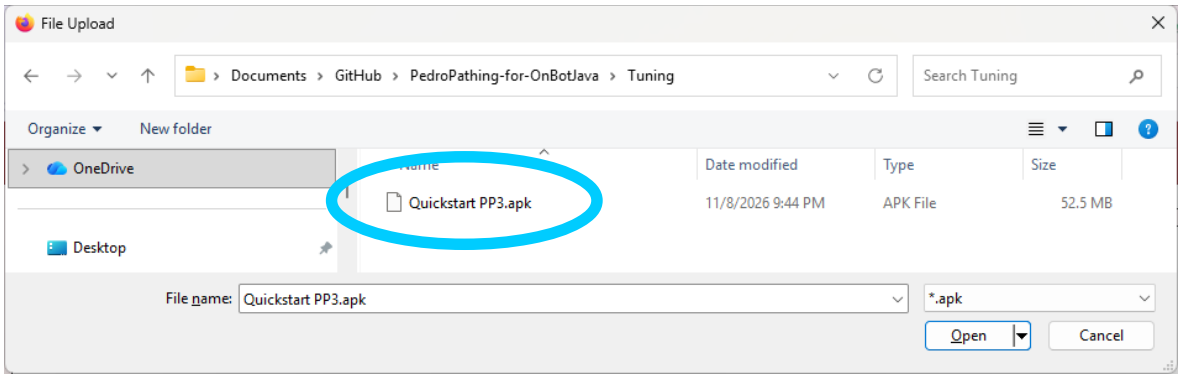
1. Go to the Manage tab on the Control Hub:



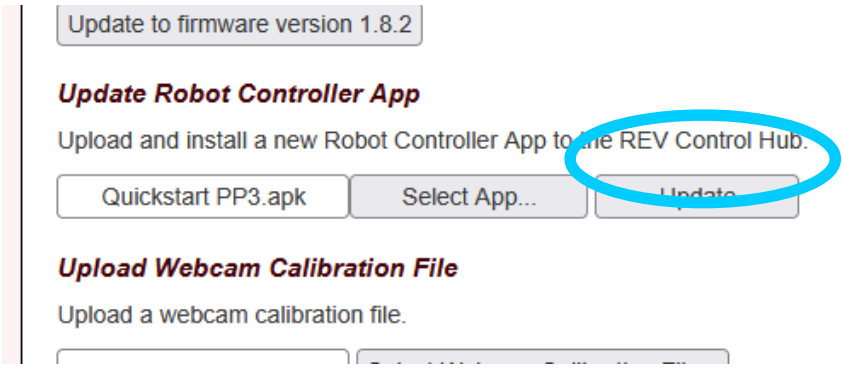
2. Go to the Upload Robot Controller app section:



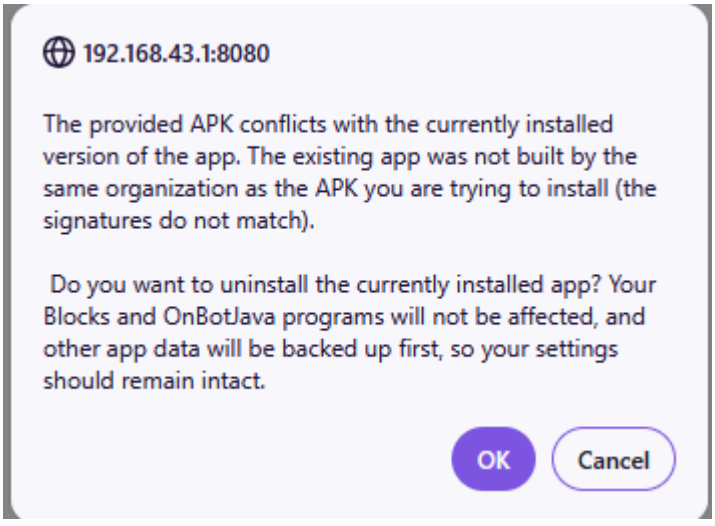
3. Click Select App and browse to the Tuning folder, select Quickstart_PP3.apk, and click Open



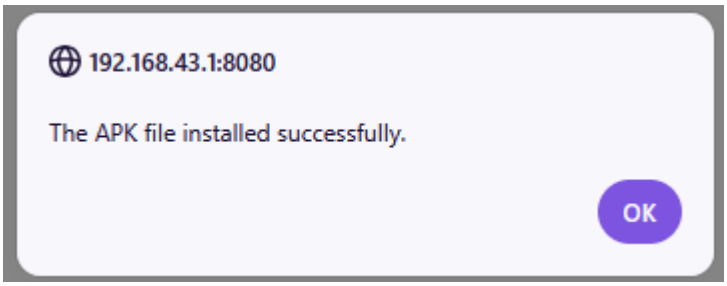
4. Select Update



5. When this message pops up, click OK

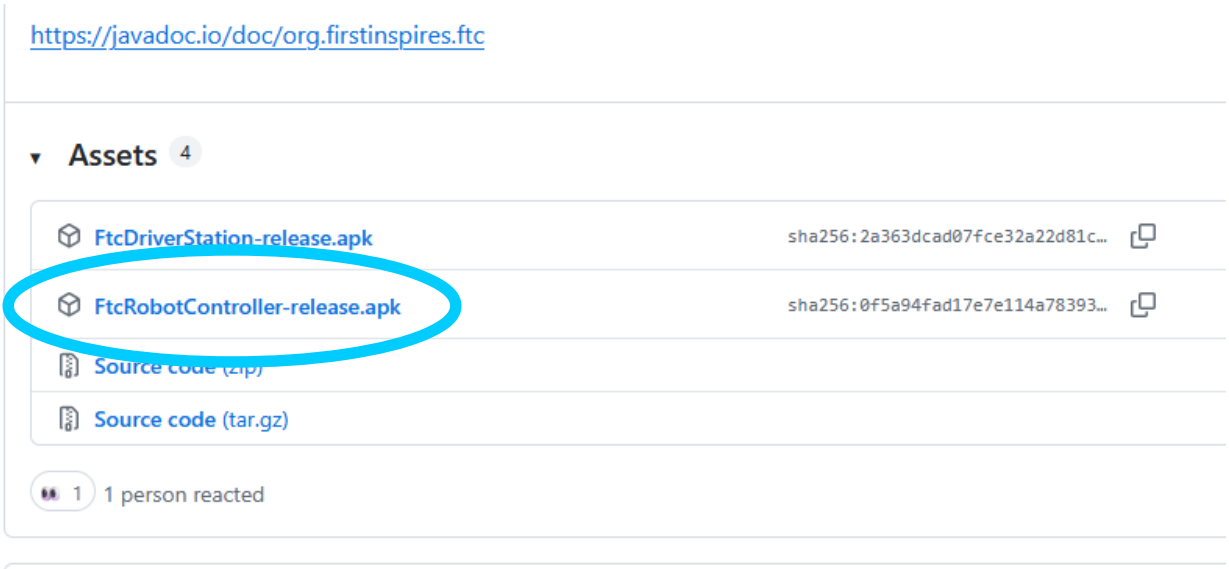


6. The Robot Controller will disconnect for a bit, then pop up this message. Click OK

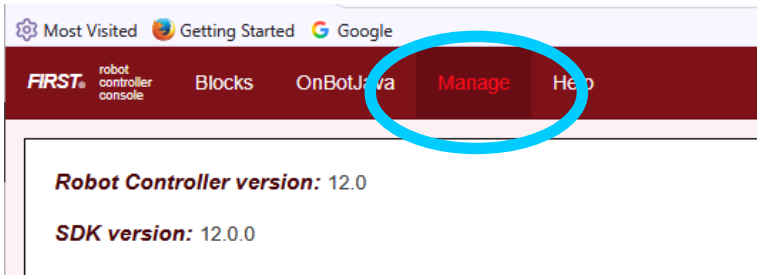


Loading FTC Control Hub app on the Contol Hub using Web Page

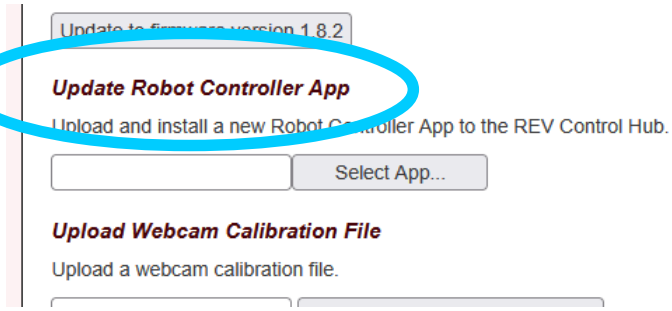
1. Download the latest FTC app from https://github.com/FIRST-Tech-Challenge/FtcRobotController/releases by clicking on the file



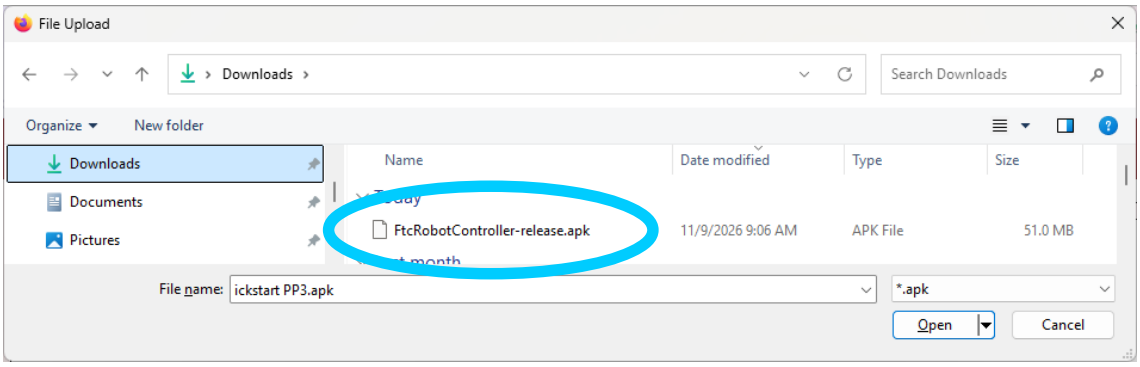
2. Go to the Manage tab on the Control Hub:



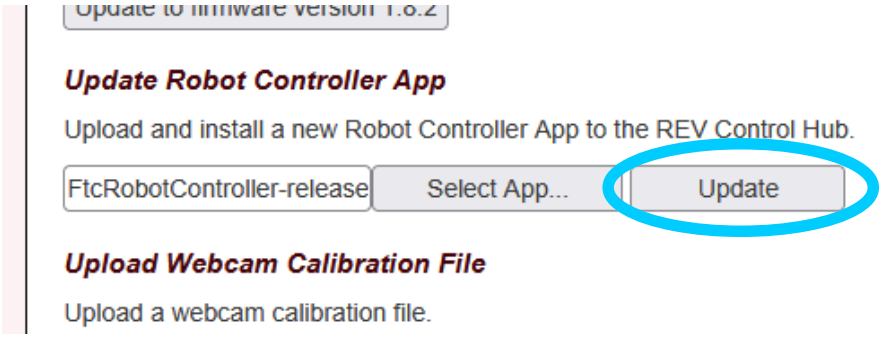
3. Go to the Upload Robot Controller app section:



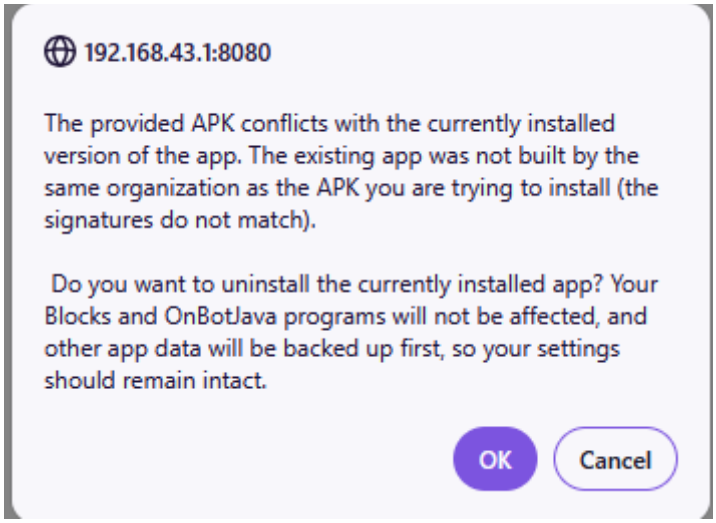
4. Click Select App and browse to the folder where you downloaded the app, select FTCRobotController-release.apk, and click Open



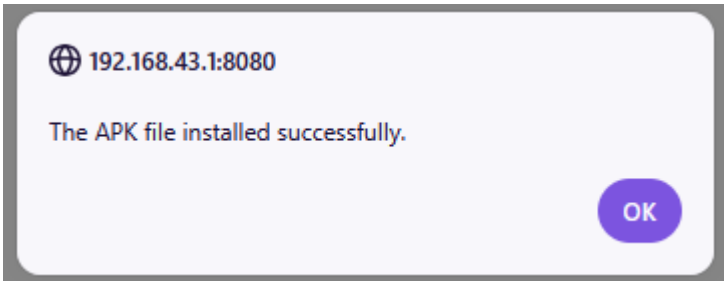
5. Select Update



6. When this message pops up, click OK



7. The Robot Controller will disconnect for a bit, then pop up this message. Click OK

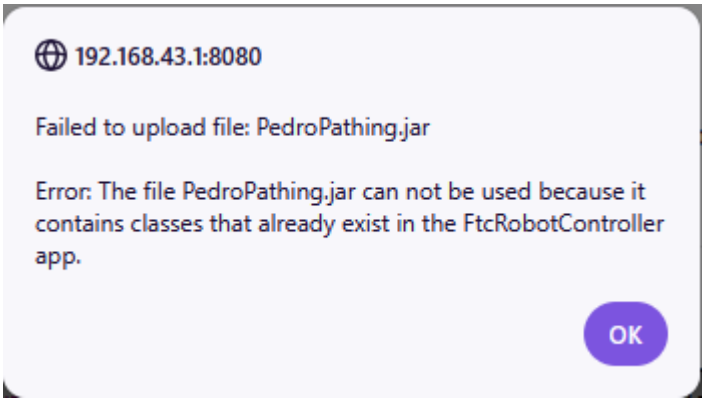


Loading FTC Control Hub app using Rev Hardware Client

These instructions are located here: https://docs.revrobotics.com/duo-control/managing-the-control-system/updating-robot-controller-application

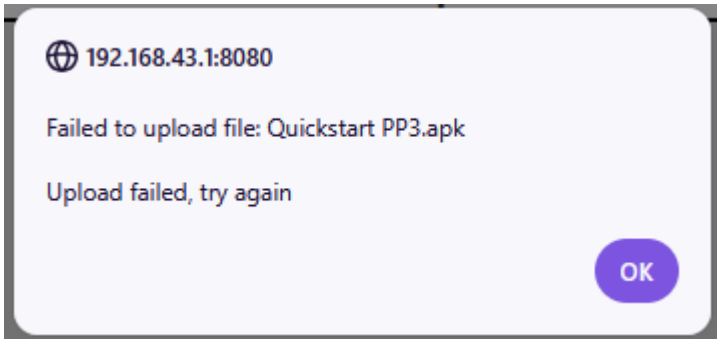
Troubleshooting

Error message:



This occurs when uploading the file with the Quickstart tuning application loaded on the robot. Load the FTC Control Hub application.

Error message:



Try again, it usually works the second time.

Error message when building OnBotJava code:

`org/firstinspires/ftc/teamcode/pedro/Constants.java line 3, column 34: ERROR: package com.pedropathing.algorithm does not exist`

Cannot build OnBotJava code when Quickstart app is loaded. Load the FTC Controller app.